

Electromagnetic Induction

Magnetic Flux:

(*) No. of $\vec{B}$ lines passing through a surface.

$$\Phi_B = \vec{B} \cdot \vec{A} = B \cdot A \cdot \cos\theta \quad (\vec{B} \rightarrow \text{uniform})$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad (\vec{B} \rightarrow \text{variable})$$

(*) SI Unit: $T \cdot m^2 = Wb$

(*) Other Units: Gauss. $cm^2 = \text{Maxwell}$
 $1 \text{ Maxwell} = 10^{-8} \text{ Wb}$

Volt. sec

$$[\Phi_B] = M \cdot L^2 \cdot T^{-2} \cdot A^{-1}$$

(*) Flux in the plane of $\vec{B}$ is zero.



$$\Phi_B = 0$$

Ex: Find flux through ring if $(l \gg R)$

Sol: R must be very small.

So,

$$\Phi_{\text{centre}} = \frac{2\sqrt{2} \mu_0 I}{\pi l}$$

$$\Phi = \frac{2\sqrt{2} \mu_0 I R^2}{l}$$

Ex: Find flux through square loop.

$$d\Phi = \vec{B} \cdot d\vec{A} \cdot \cos 180^\circ$$

$$\Phi = \int_a^{a+l} \frac{\mu_0 I}{2\pi x} \cdot l \cdot dx$$

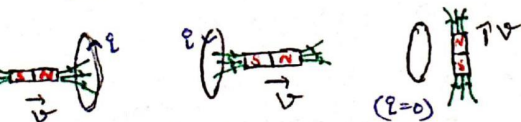
$$\Phi = \frac{\mu_0 I l}{2\pi} \log_e \left(\frac{a+l}{a} \right) \quad \text{Ans}$$

NOTE: Φ_B will always be in the format of $K(\mu_0 I \times \text{distance})$, where K is some constant.

Lenz's Law:

→ Based on cons. of energy

→ Dirⁿ of induced current in a wire wd. always be such that it opposes change in Φ_B .



Faraday's Law:

→ Magnitude of induced emf is equal to rate of change of magnetic flux.

→ Emf induced does not depend on material of loop (wood, plastic, metal).

→ Emf induce same hoga, par current sirf conductor mein flow hoga.

$$(emf)_{\text{inst.}} = -\frac{d\Phi}{dt}$$

$$(emf)_{\text{avg}} = -\frac{\Delta\Phi}{\Delta t}$$

$$I_{\text{inst.}} = -\frac{d\Phi}{R \cdot dt}$$

$$I_{\text{avg}} = -\frac{\Delta\Phi}{R \cdot \Delta t}$$

$$\Delta Q = \frac{\Delta\Phi}{R}$$

→ Amount of total ind. q flowing in a ckt is independent of time.

Ex: Long solenoid carry current $I = kt e^{-kt}$. A loop of $2R$ radius placed in equatorial plane of solenoid. Graph of I_{ind} .

Sol: $B = \mu_0 n I = \mu_0 n k t e^{-kt}$

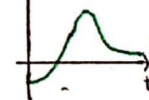
$$\Phi = B \cdot A \cdot \cos 0^\circ = 4\pi \mu_0 k R^2 t e^{-kt}$$

$$-\frac{d\Phi}{dt} = \frac{-4\pi \mu_0 k R^2 n}{B} \frac{d(kt e^{-kt})}{dt}$$

$$-\frac{d\Phi}{dt} = -B e^{-kt} [1 - kt]$$

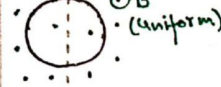
$$I_{\text{ind.}} = \frac{-B e^{-kt}}{R_0} [1 - kt]$$

For $t=0$, $I_{\text{ind.}} = \frac{-B}{R_0}$



Rotating Coil/Loop:

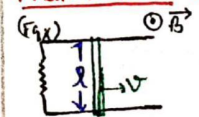
$$\omega = \frac{\theta}{t} \Rightarrow \theta = \omega t$$



$$emf = \frac{d\Phi}{dt} = BA\omega \sin(\omega t)$$

$$\text{Max. emf for } N \text{ loops} = NBA\omega$$

Rail Problem:



$$\Phi_{\text{loop}} = B \cdot (lx)$$

$$\frac{d\Phi}{dt} = B \cdot l \left(\frac{dx}{dt} \right)$$

$$emf_{\text{loop}} = Bv \cdot l \quad (\vec{v} \times \vec{B}) \cdot \vec{l}$$

→ Put right hand along $\vec{v}$ and curl fingers along $\vec{B}$, then thumb gives dirⁿ of higher potential.

$$I = \frac{Bv \cdot l}{R}$$

$$F = \frac{B^2 l^2 v}{R}$$

Force acts opp to velocity, so on Fext. is essential.

$$\text{Power dissipated} = F \cdot v = \frac{B^2 l^2 v^2}{R} \text{ by Fext.}$$

Hall effect / Motional emf:

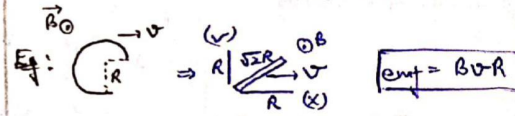
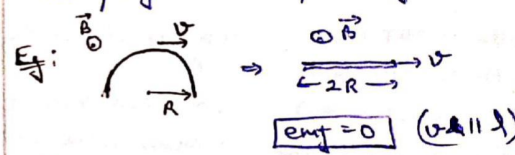
→ According to Lenz & Faraday, emf is induced in loop.

→ But, Hall said that emf was induced in conducting rod due to motion in magnetic field.

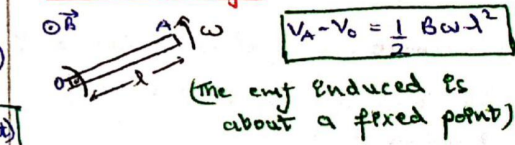
For emf to be induced, $\vec{v}$, $\vec{l}$ & $\vec{B}$, all must be mutually $\perp$ to each other.

$$emf = (\vec{v} \times \vec{B}) \cdot \vec{l}$$

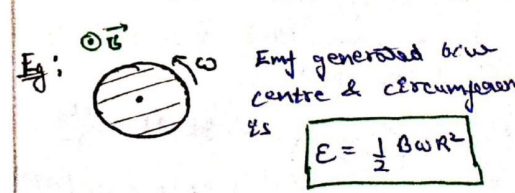
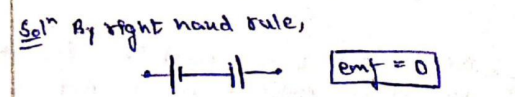
→ For a non-linear conductor, take its projection for emf calculation.



Rotational emf:

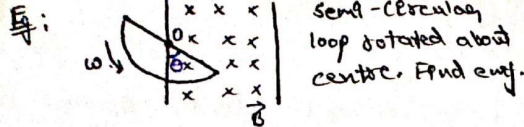


$$V_A - V_B = \frac{1}{2} B \omega l^2$$



$$E = \frac{1}{2} B \omega R^2$$

Find V_{AC} . $\frac{3}{8} B \omega l^2$



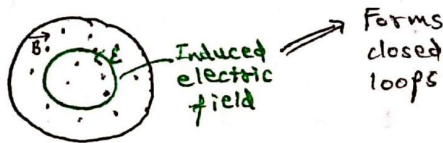
Solⁿ Area = $\frac{1}{2} \times \text{Radius} \times \text{Arc}$

So, $\phi = B \times \left(\frac{1}{2} \times R \times R\theta\right) \Rightarrow \frac{BR^2\theta}{2}$

$\frac{d\phi}{dt} = \mathcal{E} = \frac{BR^2}{2} \frac{d\theta}{dt} \Rightarrow \frac{1}{2} BR^2 \omega$

Induced electric field:

→ When a cylindrical, symmetric time varying magnetic field is present in the region. Then, a concentric circular $\vec{E}$ is induced.



Electric Field

(अवकल) (अवकली)

- Due to rest charge
- Electrostatic field
- Conservative field
- Never form closed loop
- $\oint \vec{E} \cdot d\vec{r} = 0$

- Due to time varying $\vec{B}$
- Non-conservative
- Always form closed loop
- $\oint \vec{E} \cdot d\vec{r} = \frac{d\phi_B}{dt}$
- $= A \cdot \left(\frac{dB}{dt}\right)$

अतः $\vec{B}$ change hogi, utni strong $\vec{E}$ induce hogi

NOTE: space varying $\vec{B}$ never induce $\vec{E}$.

3) Here $\vec{B}$ is inducing field & $\vec{E}$ is induced field

4) Induced & inducing field are always far to each other.

Q: there exist a mag. field in circular region of radii R as $B = B_0 t$, far to plane. Find $\vec{E}_{ind}$ inside & outside circle.

Solⁿ (9) $\oint \vec{E} \cdot d\vec{l} = \frac{d\phi_B}{dt}$
 $E \phi 2\pi r = \pi r^2 \left(\frac{dB}{dt}\right)$

$\Rightarrow E \cdot 2\pi r = \pi r^2 \left(\frac{dB}{dt}\right)$

$\Rightarrow E_{in} = \frac{B_0 r}{2}$ $\therefore E_{in} \propto r$

(99) $E \cdot \phi 2\pi r = \pi r^2 \left(\frac{dB}{dt}\right)$
 $E \cdot 2\pi r = \pi r^2 \frac{B_0}{2\tau}$
 $E = \frac{B_0 R^2}{2\tau}$

$\therefore E_{out} \propto \frac{1}{r}$

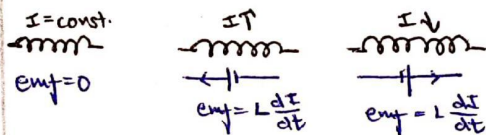
Self Induction & solenoid:

→ If the current passing through a solenoid is changed, then solenoid itself induce an opp. current.



$\phi = \mu_0 n^2 A l I$ $\frac{d\phi}{dt} = |emf| = L \frac{dI}{dt}$

$[L = \text{self inductance} = \mu_0 n^2 A l]$



→ Since,

$L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}$

If $n = \text{const} \Rightarrow L \propto l$

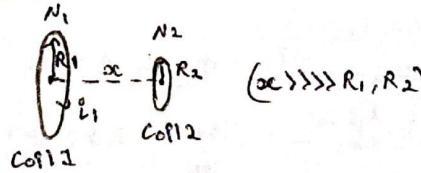
If $N = \text{const} \Rightarrow L \propto 1/l$

Energy stored in solenoid $= E = \frac{1}{2} L I^2$

Energy density $= U = \frac{B^2}{2\mu_0}$

Mutual Inductance:

Mutual inductance is induction of current in a pair of coils due to each other's current.



Flux in coil-2 due to coil-1

$\phi_{21} = B_1 A_2$
 $= \frac{\mu_0 I_1 R_1^2 N_1}{2(R_1^2 + x^2)^{3/2}} \cdot \pi R_2^2 \cdot N_2$

$\phi_{21} = \left(\frac{\mu_0 \pi R_1^2 R_2^2 N_1 N_2}{2x^3}\right) I_1$

$\phi_{12} = \left(\frac{\mu_0 \pi R_1 R_2 N_1 N_2}{2x^3}\right) I_2$

→ The constant quantity is called 'Mutual Inductance' and is equal for both coils.

$\therefore \phi_{21} = M_{21} \cdot I_1$ $\phi_{12} = M_{12} \cdot I_2$

$M_{21} = M_{12}$

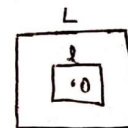
(Reciprocity Theorem)

→ Induced emf in coils

$(emf)_{\text{coil-2}} = M_{21} \cdot \frac{dI_1}{dt}$

$(emf)_{\text{coil-1}} = M_{12} \cdot \frac{dI_2}{dt}$

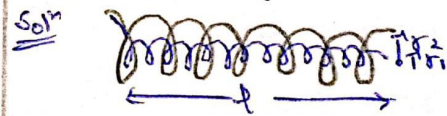
Eg: Calculate mutual inductance of two concentric coils of side length l & l ($l \gg l$).



Solⁿ $\phi_{12} = M_{12} \cdot I_1$
 $\Rightarrow \left(\frac{2\sqrt{2}\mu_0 I_1}{\pi l}\right) l^2 = M_{12} \cdot I_1$

$M_{12} = \frac{2\sqrt{2}\mu_0 l^2}{\pi l}$ Ans

Eg: Find ratio of L & M of two coaxial solenoids of radii r_1 & r_2 and turns per length n_1 & n_2 .



$L = \mu_0 n_1^2 l \pi r_1^2$

$M = \phi_1 / I_1 = \mu_0 n_2 I_1 (\pi r_1^2 n_1 l)$

$\Rightarrow M = \mu_0 n_2 n_1 \pi r_1^2 l$

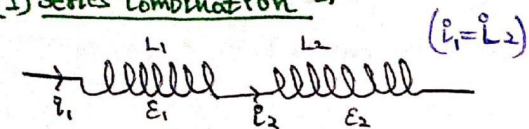
So $\frac{M}{L} = \frac{\mu_0 n_1 n_2 \pi r_1^2 l}{\mu_0 n_1^2 l \pi r_1^2}$

$\Rightarrow \frac{M}{L} = \frac{n_2}{n_1}$ $\left[\frac{L}{M} = \frac{n_1}{n_2} \right] \text{ Ans}$

Electromagnetic Induction-2

Combination of Induction:

I) Series Combination $\Rightarrow$



When, magnetic flux of coil 1 does not link with coil 2 ($K=0$), as ideal condⁿ, then

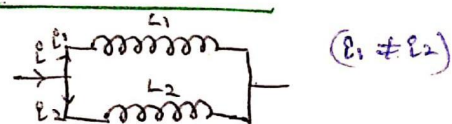
$$L_{eq} = L_1 + L_2$$

If ($K \neq 0$) Φ_2 of coils is linked together, then

$$L_{eq} = L_1 + L_2 \pm 2M$$

+2M if connected in same sense
-2M if connected in opp. sense

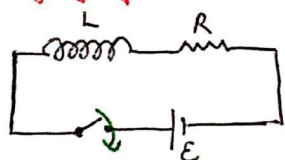
II) Parallel Combination $\Rightarrow$



Here,

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$$

Charging of Inductor:



At $t=0$

Inductor does not allow current to flow, developing an equal & opp. voltage across to battery and behave as open wire.

$$L \cdot \frac{dI}{dt} = E \quad (I=0)$$

At $t=\infty$, steady state

At steady state, inductor behaves as a normal wire and current flowing is maximum.

$$I_{max} = E/R$$

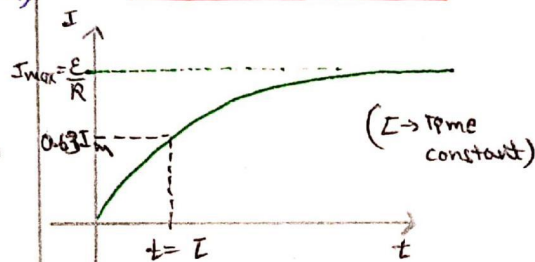
$\rightarrow$ All the potential difference is across the resistance R.

$$V_L = 0$$

$$V_R = E$$

The function of current in circuit is given by,

$$I = I_0 \left(1 - e^{-\frac{tR}{L}}\right)$$



Decay of current in the LR circuit:

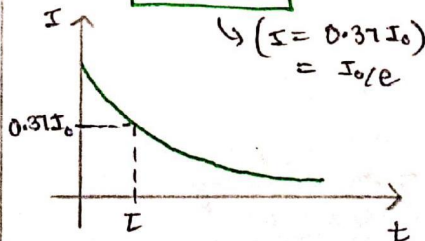
The eqⁿ of current during decay is given by,

$$I = I_0 e^{-t/\tau}$$

where, $I_0 \rightarrow$ Initial current
 $\tau \rightarrow$ Time constant

Also,

$$\tau = L/R$$

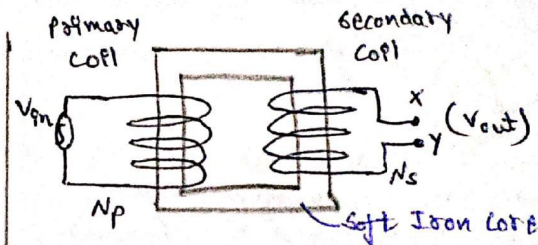


Transformer:

$\rightarrow$ Based on the mutual inductance of two coils.

$\rightarrow$ Depending upon regulated output voltage.

- $\rightarrow$ Step-up transformer
- $\rightarrow$ Step-down transformer



$\rightarrow$ Magnetic flux linked with both coils is same.

$$(i) \quad \frac{V_{in}}{V_{out}} = \frac{N_p}{N_s} \quad (V \propto \text{No. of turns})$$

$N_p \rightarrow$ No. of turns in Primary coil

$N_s \rightarrow$ No. of turns in Secondary coil

$$(ii) \quad \frac{I_{in}}{I_{out}} = \frac{N_s}{N_p} \quad \left[I \propto \frac{1}{\text{No. of turns}} \right]$$

(iii) Generally, we assume an ideal transformer with zero power loss. But if non-ideal transformer, then

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{out} \cdot I_{out}}{V_{in} \cdot I_{in}}$$

Step-up Transformer

$$N_s > N_p$$

$$V_{out} > V_{in}$$

$$I_s < I_p$$

$$P_{in} = P_{out}$$

Step-Down Transformer

$$N_s < N_p$$

$$V_{out} < V_{in}$$

$$I_s > I_p$$

$$P_{in} = P_{out}$$

Eddy Current in Time Varying Magnetic Field:

$\rightarrow$ These are miniature localized currents in a metal body.

$\rightarrow$ When a metal body is placed in time varying magnetic field, the free e^- present in it rotate such that the external MF is opposed (Lenz law).

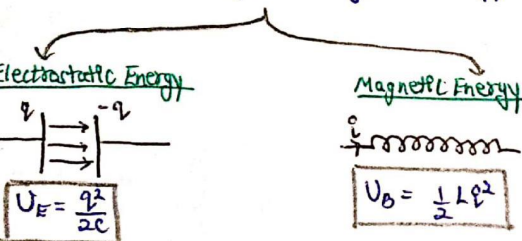
$\rightarrow$ Due to this, miniature currents called 'Eddy Current' are developed throughout the body.

- ⑤ B.T.
- * Due to eddy current, bodies continuously radiate heat.
 - * Microwave also works on this phenomenon.

Thus, if any body containing free e^- is placed in time varying MF, then it develops eddy current & radiates heat.

LC Oscillations:

→ In a circuit consisting inductor L & capacitor C (Assuming no resistance in ckt at all), there exist oscillation b/w electrostatic energy and magnetic energy.



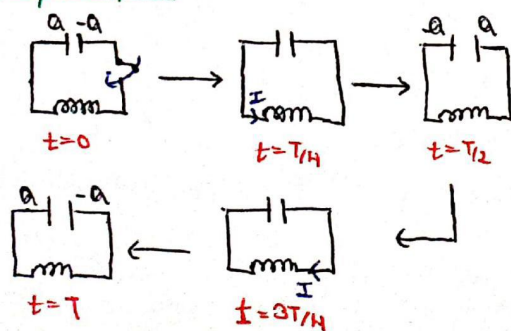
→ Consider, initially the capacitor plates are charged by a charge.

→ When switch is closed, then charge starts flowing from +ve plate until entire charge becomes zero and entire electrostatic energy converted into magnetic energy.

→ After this, the charge start developing on the capacitor plates again, but in opp. polarity.

→ This process keeps going and U_E & U_B oscillates this way.

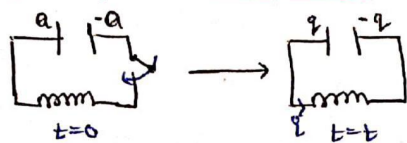
Complete Cycle ⇒



* Time taken for U_E to convert into U_B completely = $T/4$

* Time taken for U_B to completely convert into U_E again = $T/4$

Charge on Capacitor at any time t ⇒



At any instant, sum of U_E & U_B is always constant. So at $t=t$

$$\frac{q^2}{2C} + \frac{1}{2} L I^2 = \text{const.}$$

$$\frac{q}{C} \frac{dq}{dt} + L I \frac{dI}{dt} = 0$$

$$\frac{q}{C} + L \frac{d^2 q}{dt^2} = 0 \quad \left(\because I = \frac{dq}{dt} \right)$$

$$\Rightarrow \frac{d^2 q}{dt^2} + \frac{q}{LC} = 0$$

For an SHM, the double differential eqn is given by

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

On comparing with above eqn

$$\omega = \frac{1}{\sqrt{LC}} \Rightarrow T = 2\pi\sqrt{LC}$$

$$\Rightarrow q = Q \sin(\omega t + \phi)$$

$Q \rightarrow$ Max. charge on capacitor

$\omega \rightarrow$ Angular freq. of LC oscillation

$\phi \rightarrow$ Initial phase of oscillation

To calculate ϕ , substitute value of charge on capacitor @ $t=0$ in $q = Q \sin(\omega t + \phi)$ and solve for ϕ .

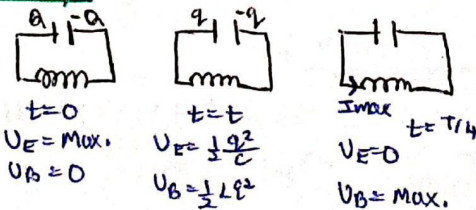
$$\text{Eg: } q = Q \sin(\omega t + 30^\circ)$$

$$\text{So}^n \text{ At } t=0, q = Q \sin 30^\circ$$

$$q = Q/2$$

Means, at $t=0$, capacitor plates have half of the max. charge on them.

Summary ⇒



⇒ The energy in ckt is constant throughout the cycle

$$\frac{Q^2}{2C} = \frac{1}{2} L I_{\text{max}}^2 = \frac{1}{2} L I^2 + \frac{q^2}{2C}$$

$$2) \quad q = Q \sin(\omega t + \phi), \quad \omega = \frac{1}{\sqrt{LC}}$$

3) Complete U_E convert into U_B and vice-versa in $(T/4)$ time

4) U_E to U_E or U_B to U_B convert in $(T/2)$